Usman Khalid

Senior Software Engineer | Technical Lead | Distributed Systems & Applied AI
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SUMMARY

Senior Software Engineer and Technical Lead with 9+ years designing scalable, fault-tolerant distributed systems in Node.js, Laravel and Python, and leading cross-functional teams. Spearheaded a multi-hundred-gigabyte database migration that deprecated MongoDB entirely and cut thousands of dollars a month in cloud costs, overhauled CI/CD and observability to hold a 99% uptime SLA, and built applied-AI tooling that accelerated PR approvals by 35% while halving manual QA time. Ships end to end, from event-driven backends to React and React Native front ends, and leads by mentoring, reviewing and setting technical direction rather than by title.

WORK EXPERIENCE

Senior Full-Stack Engineer / Technical Lead
Feb 2024 - Apr 2026

Premion (TEGNA) | California, USA (Remote)

- Architected and maintained high-scale Node.js microservices (Express, Fastify) powering an industry-leading CTV/OTT advertising platform running high-volume ad campaigns.
- Delivered attribution systems end to end across the Forecast, Creative and Attribution modules, exposing Swagger-documented REST APIs consumed across the platform.
- Designed the real-time analytics storage strategy across MongoDB, PostgreSQL and Snowflake; drove the migration off MongoDB as data grew into hundreds of gigabytes, reducing cost and improving analytical join performance.
- Held system uptime through significant post-acquisition data growth via SQL query optimization, indexing, and AWS and Docker infrastructure tuning.
- Led code review and approval for junior developers; set technical direction on schema and module design, and mentored the team on performance patterns such as eliminating N+1 queries.
- Owned sprint scope and delivery in coordination with the CEO and product manager; tracked work in Jira and versioned in Git.
- Engineered Prism PR Copilot, an AI-driven CI/CD tool running Claude on AWS Bedrock that validates Jira acceptance criteria against pull requests and posts its review to GitHub, accelerating PR approvals by 35% and saving 50% of manual QA time.

Senior Full-Stack Engineer
Apr 2022 - Feb 2024

Octillion | California, USA (Remote), acquired by Premion

- Designed PostgreSQL and Snowflake schemas carrying high-throughput CTV/OTT advertising data at platform scale.
- Built Node.js APIs and migrated the services behind them onto Docker containers on AWS, making deployments consistent across every environment.
- Owned the database layer end to end (schema design, indexing and query optimization) as advertising data volume outgrew the original architecture.

Engineer
May 2020 - May 2022

Sprout Tech | Remote

- Launched Content Gorilla from scratch and scaled it to 20,000+ active users, building and maintaining the full stack on Laravel and Vue.js.
- Built an automated AI content engine that processed 178M+ words and 500k+ blog posts, integrated with the WordPress and Medium REST APIs.
- Automated video-to-blog conversion and social scheduling, keeping delivery reliable at scale by wiring third-party publishing APIs and webhooks into the platform.

Engineer
Jan 2017 - May 2020

Outstart | Lahore, Pakistan

- Engineered a multi-tenant property management platform for New York-based owners, automating maintenance requests and apartment service workflows end to end.
- Delivered LePark, a Laravel parking management system synchronizing in real time with field parking hardware (IoT), with secure REST APIs bridging web dashboards and on-site infrastructure.

PROJECTS

Premion DSP: Real-Time Bidding Platform (Demand-Side)
Node.js, Redis, Snowflake, OpenRTB, CTV/OTT

- High-throughput real-time bidding engine processing OpenRTB bid requests for CTV/OTT inventory: audience targeting, dynamic bid valuation and automated campaign budget pacing.
- Architected low-latency Node.js and Redis microservices holding sub-100ms bid responses under ad-exchange SLA, and owned the big-data telemetry pipeline into Snowflake.

Premion Fate System: Ad-Tech Reporting & Campaign Management
Vue.js, NestJS, PostgreSQL, Snowflake

- Enterprise reporting and campaign management for high-volume OTT/CTV analytics: location, sale and web attribution, creative uploads and inventory forecasting.
- Owned the system architecture, set the API documentation standards the squads worked to, and directed frontend and backend squads through structured Agile delivery.

GHL Webhook Sync Service: Event-Driven CRM Integration (Sole Engineer, Client Project)
Python 3.12, Django, Django REST Framework, Celery, Redis, PostgreSQL, Docker, Gunicorn, Nginx, AWS EC2

- Receives GoHighLevel opportunity webhooks and stores each event exactly once in PostgreSQL, then syncs it to Trello through Celery tasks with retry and exponential backoff and a tracked status lifecycle.
- Shipped with keyless IAM OIDC CI/CD, human-approved production releases, a separate staging environment and 55 automated tests.

SophieFlow: AI Social Media Management SaaS (Founder / Sole Engineer)
Laravel 12, Vue 3 + TypeScript, Inertia.js, Redis, Stripe, Google Gemini / OpenRouter, OpenAI

- Multi-tenant SaaS to schedule and publish across 12+ social networks with a unified inbox, analytics, an AI assistant and CRM-style lead capture; queued publishing with retry/backoff, 10+ OAuth integrations, Stripe billing and plan-based access.
- Built a Gemini 2.5 Pro content engine that writes and publishes SEO articles and prompt templates on schedule, and the first-generation platform (Laravel 9, OpenAI) with team workspaces and WordPress publishing.

OneParacon: Construction Management Platform, Web & Mobile (Sole Engineer, Client Project)
Laravel 13, React 18 + TypeScript, PWA, MySQL, Anthropic Claude API, Expo / React Native, SQLite

- Multi-tenant PWA with QR site sign-on (GPS geofencing, offline queue), site diaries, defects, a Gantt programme and finance, plus a Claude tool-calling assistant with role-scoped tools and immutable audit logs; 480+ API tests.
- Companion iOS/Android app in Expo with an offline-first SQLite sign-on queue, server-driven capability permissions and QR sign-in from a desktop session.

Resume Submitter: AI Job-Application Automation, Web & Mobile (Sole Engineer)
Laravel 13, Vue 3, Inertia.js, Anthropic Claude API, Gmail API, Crawlee / Playwright, Chrome Extension (MV3), Expo

- Tailors resumes to job posts from a master profile, writes cover letters and emails applications via Gmail with the PDF attached; ATS-readability checks and prompt-injection guards.
- Adaptive Crawlee/Playwright job crawler, a Manifest V3 auto-fill extension and an Expo mobile app; Docker deploys to Railway and 650+ automated tests.

SKILLS

Languages: TypeScript, JavaScript (ES6+), PHP, Python, SQL

Backend: Node.js (Express, Fastify), NestJS, Laravel, Django, Django REST Framework, Celery, Microservices

Frontend & Mobile: Vue.js (Vue 3), React, Next.js, Inertia.js, Tailwind CSS, PWA, React Native, Expo, TanStack Query, Zustand

Databases & Data: PostgreSQL, MySQL, MongoDB, Snowflake, Redis, SQLite, Query Optimization, ETL Migration

AI & LLM: Anthropic Claude API, Claude on AWS Bedrock, Tool Calling, Google Gemini, OpenAI, OpenRouter

Cloud & DevOps: AWS (EC2, RDS, S3, SSM, IAM OIDC, ELB), Docker, Nginx, Gunicorn, PM2, Railway, GitHub Actions, Bitbucket Pipelines

Architecture: System Design, OpenRTB / Real-Time Bidding, Event-Driven Webhooks, Idempotency & Retries, Big Data Pipelines, Multi-Tenant SaaS, Cost Optimization, RFC Writing

Quality & Leadership: TDD, PHPUnit, Jest, Automated QA, Code Review & Approval, Technical Direction, Mentoring, Sprint Planning, Agile/Scrum

EDUCATION

University of Lahore 2013 - 2017
B.S. Computer Science